H2 Math Summary

Notes

Maximus Pung

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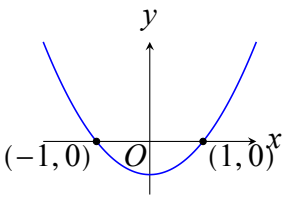
1 Graphing Techniques

1.1 Curve Sketching

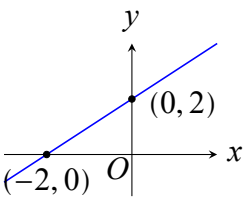
Key features of a curve:

(a) **Axial intercepts**, when $x=0$ or $y=0$.

x-intercept: set $y=0$.
e.g. $y=x^2-1$: $x^2-1=0 \Rightarrow x=\pm 1$.

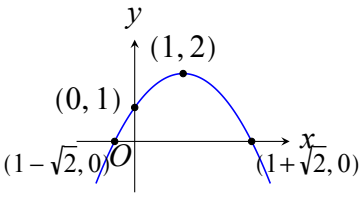


y-intercept: set $x=0$.
e.g. $y=x+2$: $y=2$.

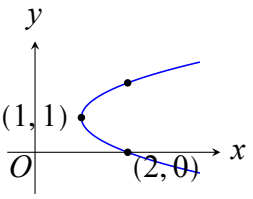


(b) **Stationary points**, when $\frac{dy}{dx}=0$ or $\frac{dy}{dx}$ is undefined. Label upon request.

Horizontal turning point: $\frac{dy}{dx}=0$ (tangent horizontal).
e.g. $y=-(x-1)^2+2$, maximum point $(1,2)$.



Vertical turning point: $\frac{dy}{dx}$ undefined (tangent vertical).
e.g. $x=(y-1)^2+1$, turning point $(1,1)$.



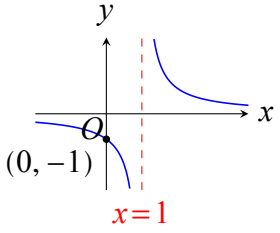
(c) **Asymptotes**

1.1.1 Vertical Asymptotes

If as $x \to k$, $y \to \pm\infty$, then $x=k$ is a vertical asymptote. Common occurrences:

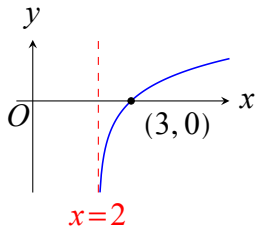
1. Fractions. Denominator $=0$.

e.g. $y=\frac{1}{x-1}$: vertical asymptote $x=1$.



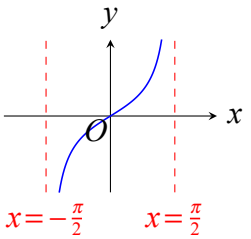
2. Logarithmic graphs. Inside $\ln = 0$.

e.g. $y = \ln(x-2)$: vertical asymptote $x = 2$.



3. Tan graphs. Inside $\tan = \frac{\pi}{2} + k\pi,\ k \in \mathbb{Z}$.

e.g. $y = \tan x$: vertical asymptotes $x = \pm\frac{\pi}{2},\ \pm\frac{3\pi}{2}, \ldots$



1.1.2 Horizontal Asymptotes

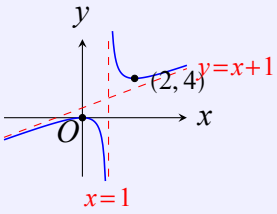
As $x \to \pm\infty$, if $y \to k$, then $y = k$ is a horizontal asymptote.

- To find horizontal asymptotes, one should sub in $x \to \pm\infty$ into the equation. This is the only foolproof way to do it.
- A curve **CAN** intersect its horizontal asymptote.

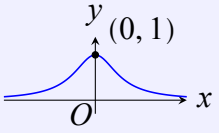
Useful trick to find horizontal asymptotes

For rational fns $\frac{P(x)}{Q(x)}$.

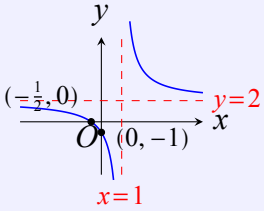
• numerator power > denom power
⇒ no horizontal asymptote
e.g. $y = \frac{x^2}{x-1}$



• numerator power < denom power
⇒ horiz asymptote: $y = 0$
e.g. $y = \frac{1}{x^2+1}$



• numerator power = denom power
⇒ horiz asymptote: ratio of coeff of highest power.
e.g. $y = \frac{2x+1}{x-1},\ y = 2$



1.1.3 Oblique Asymptotes

An asymptote of the form $y=mx+c$, $m\neq 0$.

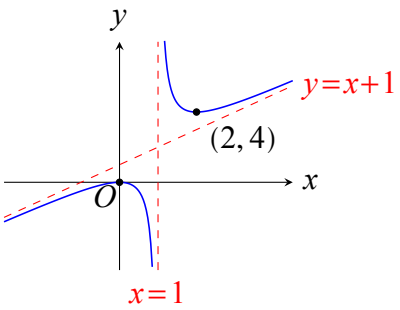
e.g. $y=2x+1+\dfrac{3}{x-2}$, $y=3x-5+\mathrm{e}^{-x}$

> **Note**
>
> For parametric curves, oblique asymptotes (if they exist) can only be found via the cartesian equation.

$$y=\frac{ax^2+bx+c}{dx+e}=Ax+B+\frac{C}{dx+e}$$

Oblique asymptote: $y=Ax+B$. Vertical asymptote: $x=-\dfrac{e}{d}$.

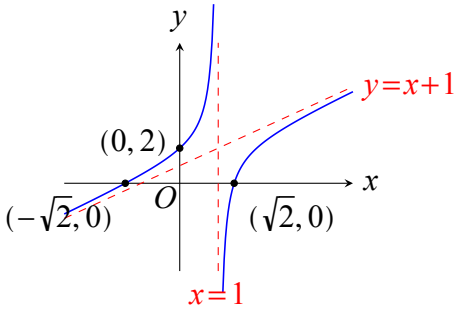
2 turning points



e.g. $y=x+1+\dfrac{1}{x-1}$

$\left(\dfrac{Cd}{A}>0\right)$

No turning points.



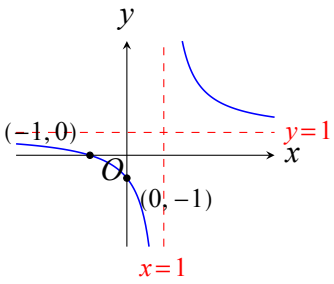
e.g. $y=x+1-\dfrac{1}{x-1}$

$\left(\dfrac{Cd}{A}<0\right)$

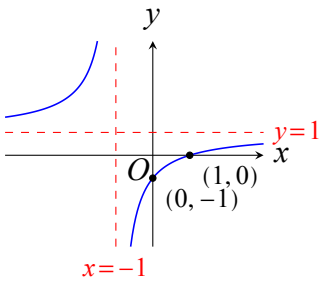
1.2 Graphs of Rational Functions

Two common shapes for each type.

Linear / linear: $y=\dfrac{Ax+B}{Cx+D}$ Horizontal asymptote: $y=\dfrac{A}{C}$. Vertical asymptote: $x=-\dfrac{D}{C}$.

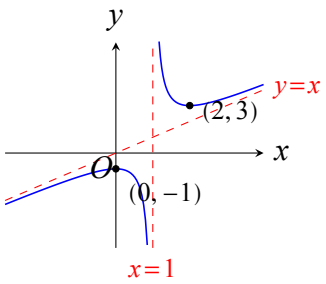


e.g. $y=\dfrac{x+1}{x-1}$

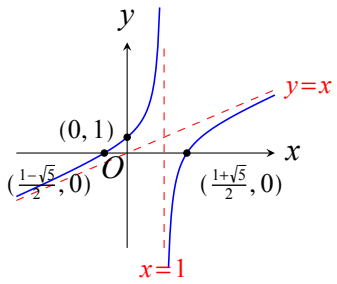


e.g. $y=\dfrac{x-1}{x+1}$

Quadratic / linear: $y=\dfrac{Ax^2+Bx+C}{Dx+E}$ Oblique asymptote; vertical asymptote $x=-\dfrac{E}{D}$.

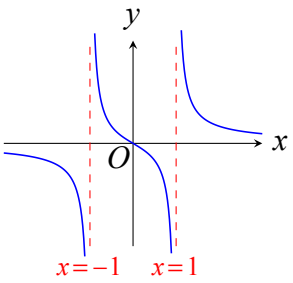


e.g. $y=x+\dfrac{1}{x-1}$

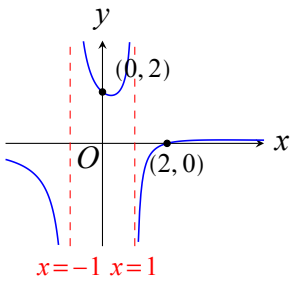


e.g. $y=x-\dfrac{1}{x-1}$

Linear / quadratic: $y=\dfrac{ax+b}{cx^2+dx+e}$ Horizontal asymptote: $y=0$. Vertical asymptotes: Solve $cx^2+dx+e=0$ (possibly no roots).



e.g. $y=\dfrac{x}{(x-1)(x+1)}$



e.g. $y=\dfrac{x-2}{(x-1)(x+1)}$

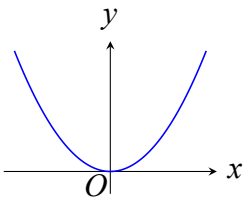
Quadratic curves: $y=a(x-h)^2+k$, $y=\alpha(x-m)(x-n)$.

1.3 Symmetry

Even functions: $\mathrm{f}(-x)=\mathrm{f}(x)$.

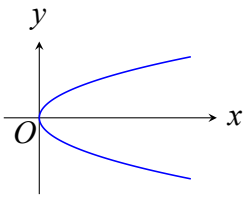
The graph is symmetric about the y-axis.

e.g. $y=x^2$



Symmetry about the x-axis: replacing y by $-y$ gives the same equation.

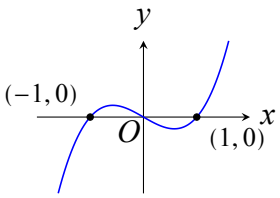
e.g. $y^2=x$



Odd functions: $\mathrm{f}(-x)=-\mathrm{f}(x)$.

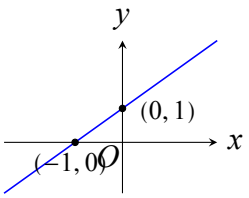
The graph is symmetric about the origin.

e.g. $y=x^3-x$

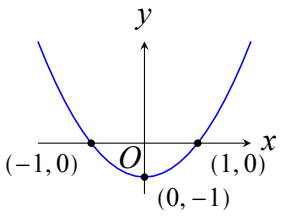


1.4 Basic Graphs

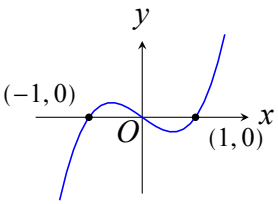
- Polynomial (linear, quadratic, cubic).
- Exponential. $y=ae^x+b$.
- Logarithm. $y=\ln(ax+b)$.
- Trigo. $y=\sin x$, $\cos x$, $\tan x$.
- Inverse Trigo. $y=\sin^{-1}x$, $\cos^{-1}x$, $\tan^{-1}x$.



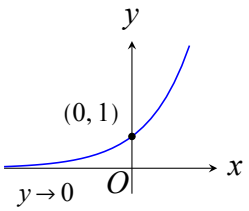
Linear $y=x+1$



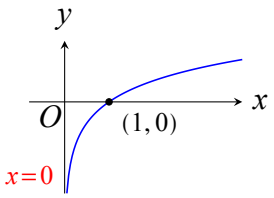
Quadratic $y=x^2-1$



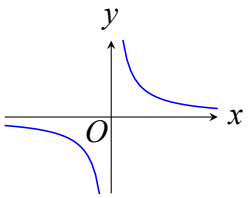
Cubic $y=x^3-x$



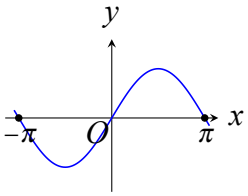
Exponential $y=e^x$



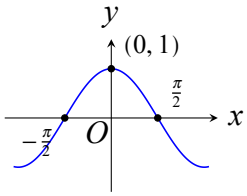
Logarithm $y=\ln x$



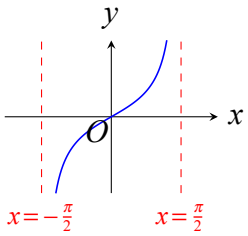
Reciprocal $y=\frac{1}{x}$



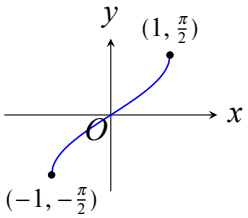
$y=\sin x$



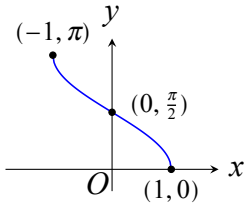
$y=\cos x$



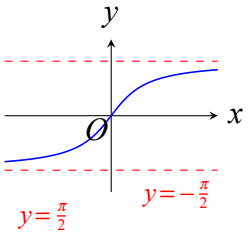
$y=\tan x$



$y=\sin^{-1}x$



$y=\cos^{-1}x$



$y=\tan^{-1}x$

1.5 Parametric Equations

$$x=x(t), \quad y=y(t), \quad t\in T$$

→ ALWAYS take note of range of t when sketching graphs.

→ Cartesian eqns. An equation of C without parameter t.

Common techniques

① Make t the subject. e.g. $x=2t-1, y=3t^2+e^t, t\in\mathbb{R}$

② Trigo Identity formulas e.g. $x=2\cos t+1, y=3\sin t-2, 0\leqslant t\leqslant \pi$

③ Others e.g. $x=t-\frac{1}{t}, y=t+\frac{1}{t}, t\neq 0$

Note: Avoid finding cartesian eqn if question did not ask you to.

→ Asymptotes:

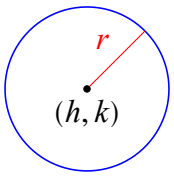
- Find (if possible) $t\to t_0$ such that $x\to x_0, y\to\pm\infty$ or $y\to y_0, x\to\pm\infty$.
- Oblique asymptotes (if they exist) can only be found via cartesian equation.

– Intersection between parametric and another curve. Sometimes both equations.

1.6 Conics

Circle: $(x-h)^2+(y-k)^2=r^2$

Centre: (h,k) radius: r

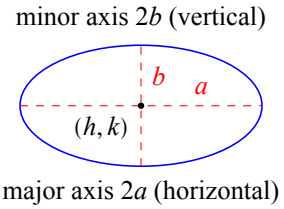


Ellipse: $\dfrac{(x-h)^2}{a^2}+\dfrac{(y-k)^2}{b^2}=1$

Centre: (h,k), "horizontal radius" $=a$, "vertical radius" $=b$

(for $a>b$: major axis horizontal, length $2a$, semi-major a; minor axis vertical, length $2b$, semi-minor b)

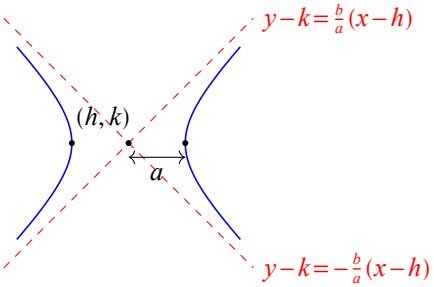
⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.



Hyperbola: Type I

$\dfrac{(x-h)^2}{a^2}-\dfrac{(y-k)^2}{b^2}=1$

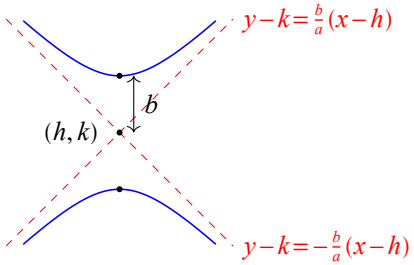
Centre: (h,k), Distance from centre to vertices $=a$



Type II

$\dfrac{(y-k)^2}{b^2}-\dfrac{(x-h)^2}{a^2}=1$

Centre: (h,k), Distance from centre to vertices $=b$



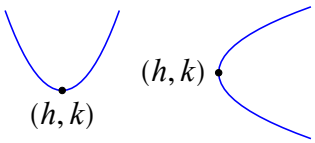
⋆ RHS ⇒ numerator coefficients of x^2, y^2 must all be 1.

⋆ Equation of asymptotes: $y-k=\pm\dfrac{b}{a}(x-h)$ (dashed in the sketches).

Parabola:

· $y-k=a(x-h)^2$ Turning point $\left(\dfrac{dy}{dx}=0\right)$: (h,k)

· $x-h=b(y-k)^2$ Turning point $\left(\dfrac{dy}{dx}\text{ undefined}\right)$: (h,k)



Common question types for curve sketching

→ Find the range/set of values that y can/cannot take. [VJC 2025]

→ Find the range of values of a such that C has no turning points [SAJC 2025]

→ Line rotation about pivot question. [SAJC 2025]

→ Conic superimposed onto graph. [DHS 2025]